

Technical Document #805: Cloud Computing - Comprehensive Overview

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Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files. Terraform and CloudFormation are popular IaC tools.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Containerization packages applications with their dependencies. Docker creates lightweight, portable containers that run consistently across environments.

Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Auto-scaling automatically adjusts compute resources based on demand. It optimizes costs while maintaining performance requirements.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Infrastructure as Code (IaC) manages infrastructure using machine-readable configuration files.
- Auto-scaling automatically adjusts compute resources based on demand.
- Multi-cloud strategies use services from multiple cloud providers.